# JACK M. EWING

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## EDUCATION

**Worcester Polytechnic Institute** — *B.S. Electrical & Computer Engineering GPA 3.66/4* **Aug 2026 – Expected May 2028**

**Lafayette College** — *Electrical & Computer Engineering (transferred to WPI) GPA 3.4 / 4* **Aug 2024 – May 2026**

**Relevant Coursework:** Embedded Design, FPGA Design, Digital Systems, Signals & Systems, Communications, Circuit Analysis, Robotics

## TECHNICAL SKILLS

**Programming:** C, C++, Python, Verilog, SystemVerilog, Java, C#, JavaScript

**Embedded & Hardware:** ESP32, Arduino, Raspberry Pi, Jetson Nano, FPGA (Artix-7 / Basys 3), I²C, BLE, GPS, IMU, PCB Design, Sensors, Data Acquisition, Soldering, Circuit Debugging

**Engineering Tools:** KiCad, LTspice, Vivado, SolidWorks, Oscilloscope, Bench Power Supply, Git/GitHub, 3D Printing, Laser Cutting

## ENGINEERING EXPERIENCE

**Formula SAE — Lafayette College** — *Embedded Systems Engineer, Telemetry & Controls* **Jan 2025 – May 2026**

- Designed, built, and tested an ESP32 Feather V2 telemetry system integrating GPS and a BNO055 IMU, capturing live position, acceleration, and orientation data at up to 15 Hz with onboard SD card logging.
- Implemented BLE and I²C communication between embedded boards to stream live vehicle data to a remote tracker; measured approximately 80 ft of reliable line-of-sight BLE range during testing.
- Designed three 2-layer PCBs in KiCad — the I/O board, the connector ("traces") board, and a combined VCU/MPR + 5 V board that consolidated two functions onto one board to reduce total board count; assembled, tested, and debugged the resulting vehicle electronics.
- Designed and tested a five-board modular safety system that commanded the ECU to shut down vehicle systems whenever the safety loop was interrupted; performed system-level integration testing across all boards.
- Debugged the 300 V high-voltage battery charging circuit using an oscilloscope, LTspice simulation, and a bench supply; worked with the Orion BMS to verify charging behavior and safe battery operation.
- Validated the telemetry system on the vehicle during driving tests, collecting clean multi-sensor data streams.

**Advanced Vertical Robotics (AVR) — Westlake Highschool**— *Team Co-Lead / Electrical & Software* **2021 – 2024**

- Co-led a nationally competitive robotics team to Top 10 overall finishes in both junior and senior seasons, leading electrical wiring, software development, and system integration.
- Developed and maintained robot software, including MQTT-based IoT communication between robotic systems and connected devices.
- Led technical brainstorming and system-development sessions while contributing to mechanical design in SolidWorks.
- Managed approximately 50% of team 3D printing, producing functional components in TPU, ABS, and PLA.

**Brain-Computer Interface Research — Lafayette College** — *Undergraduate Researcher* **Sep 2024 – May 2026**

- Collected neural-signal datasets using OpenBCI and commercial BCI hardware for wheelchair-control research, and developed Python workflows for organizing and processing the recorded data.

**Simpler Trading** — *Software Engineering Intern* **Summer 2026**

- Built a Python tool to combine and evaluate outputs from multiple trading indicators, processing S&P 500 data from 2020–2026 via the Polygon API and generating CSV datasets and analytical visualizations.

## PROJECTS

**Digital Logic Design — FPGA** — *Verilog / SystemVerilog*

- Designed, simulated, and synthesized digital logic in Verilog and SystemVerilog, implementing and verifying designs on a Basys 3 (Xilinx Artix-7) FPGA using Vivado.

## AWARDS

- National Championship, Advanced Vertical Robotics — #1 Design in the Nation, 2023
- National Championship, Advanced Vertical Robotics — #1 Presentation in the Nation, 2022
- Certificate of Recognition, State Representative Vikki Goodwin — Bell Helicopter Vertical Robotics Competition, 2021